

THEME 2 – COMPLEX NUMBERS

The earliest stage of number development concerns the so-called counting numbers

$$1, 2, 3, \dots$$

which are known as the **natural numbers** or the **positive integers**. When the sum of two natural numbers or their product is computed, the result is always a natural number. This means that the natural numbers are **closed** with respect to addition and multiplication. The equation

$$5 + x = 3$$

however, cannot be solved within the set of positive integers. The number concepts that we denote by **zero** and the **negative integers** were invented to solve such equations. By enlarging our number system to contain all integers

$$\dots -3, -2, -1, 0, 1, 2, 3, \dots$$

we obtain a set which is closed under addition, subtraction and multiplication. We all know that the set of **integers** still is not big enough, since the equation

$$mx = n$$

for example, cannot always be solved within this set. This leads to the introduction of the set of all **rational numbers** (which of course includes all integers). This set proves to be closed under addition, subtraction, multiplication and division. (Of course we cannot divide by zero!) We all know that the equation

$$x^2 = 2$$

has no solution in the set of rationals. Again we have to enlarge our set of numbers to include also the set of the so-called irrational numbers like $\sqrt{2}$, π , e , etc. Our new set of numbers is called the **real numbers**. This is the set of numbers with which you have been working in Calculus this far.

You are all aware of the fact that the equation

$$x^2 + x + 1 = 0$$

has no real solutions, since the discriminant

$$b^2 - 4ac = 1 - 4 = -3$$

is negative. Thus we have to invent an even bigger system of numbers namely the set of **complex numbers**.

These numbers will now be the object of our study.

You may well ask why we have to study these numbers. The reason is that even in fields outside of pure mathematics like engineering, these numbers seem to be very well suited to solve problems concerning damped vibrations, harmonic oscillations, alternating currents, and other wave phenomena.

UNIT 2.1 - Operations and geometric interpretation

Our first step is to define a new number i , having the property that its square equals -1 .

Next (Notes, Definition 2.1 p 47) we define a **complex number** as an object of the form $a + bi$, where a and b are real numbers.

It should now be clear that the term **complex** does not mean **complicated** but rather refers to something consisting of more than one part.

In fact, a complex number has a **real** as well as an **imaginary** part.

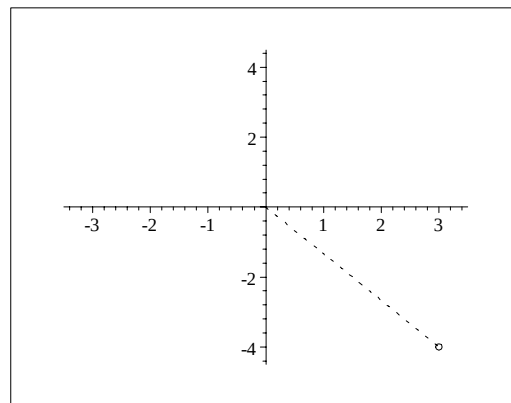
Do pay attention to the definition of equality of two complex numbers (Notes, Definition 2.1 p 47.)

Complex numbers can be represented geometrically (Notes, Paragraph 2.3 on p 48).

This will play an important role as we develop this topic.

You will be asked to draw simple graphs to indicate a specific set of complex numbers.

The **modulus** (Notes, Definition 2.4 p 48) of a complex number is the distance between the origin and the point in the complex plane.



The sketch shows the complex number $z = 3 - 4i$ and the modulus of z is the length of the dotted line between the origin and the point.

Of course, one hopes to have the same algebraic operations for complex numbers as for the real numbers. Indeed, we will learn how to add, subtract, multiply (Notes, Definition 2.6 p 48) with complex numbers. These operations have the same properties as the properties for real numbers.

Before we can divide with a non-zero complex number, you have to know what the **conjugate** of a complex number is (Notes, Definition 2.8 p 49).

One important property of the conjugate is that the product of a complex number and its conjugate is always a real number as shown on p 49.

$$\text{If } z = 2 - 4i \text{ then } z \times \overline{z} = 2 - 4i \times \overline{2 - 4i} = (2 - 4i)(2 + 4i) = 20$$

We use the property for division of complex numbers (Notes, Definition 2.9 p 50).

Example

Write the complex number $\frac{3+i}{2-4i}$ in the form $a + ib$.

$$\frac{3+i}{2-4i} = \frac{3+i}{2-4i} \times \frac{2+4i}{2+4i} = \frac{2+14i}{20} = \frac{1}{10} + \frac{7}{10}i$$

You must know all the properties of the conjugate (Notes, Theorem 2.11 p 50) and must be able to prove it. (Leave number (4) until we have done mathematical induction in Unit 3.1.)

It soon becomes clear that new possibilities arise.

You can now write the square root of a negative number as a complex number:

$$\sqrt{25} = 5, \quad -\sqrt{25} = -5 \quad \text{and} \quad \sqrt{-25} = \sqrt{25}i \quad \text{a complex number.}$$

In the field of complex numbers many equations which have no real solutions, now do have complex solutions. As a matter of fact, **every** quadratic equation in one unknown now always has two (not necessarily different) solutions. Eventually we will be able to prove a much stronger result.

If you have to solve the equation $ax^2 + bx + c = 0$, and a, b and c are **real numbers**, you can use the formula to find the two solutions.

Example:

Solve the equation $x^2 + x + 1 = 0$.

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-1 \pm \sqrt{-3}}{2} \\ &= \frac{-1 \pm \sqrt{3}i}{2} \end{aligned}$$